PSL Cricket Analytics Dashboard Project Report

1. Dataset and Data Sources

This project uses Pakistan Super League (PSL) cricket data covering the period from 2016 to 2023. The dataset consists of three primary data sources along with a manually created reference table for team standardization.

The **batting dataset** contains **521 records and 17 columns** representing player batting performance metrics such as:

- Runs scored
- Batting average
- Strike rate
- Fours and sixes
- Centuries and half-centuries
- Innings played

The **bowling dataset** contains **521 records and 17 columns** representing bowling performance statistics including:

- Wickets taken
- Economy rate
- Overs bowled
- Maiden overs
- Bowling average
- Bowling strike rate
- Best bowling figures

The **matches dataset** contains **242 records and 8 columns** representing match-level information such as:

- Match date
- Competing teams
- Match winner
- Margin of victory
- Venue
- Result type

Additionally, a **teams reference table** containing **6 records** was manually created to standardize franchise naming conventions and support relational mapping across all datasets.

Together, these datasets represent:

- Six PSL franchises
- Seven PSL seasons (2016–2023)
- More than 340 unique players
- 242 completed matches

The dataset provides sufficient historical coverage to support descriptive, predictive, and prescriptive analytics.

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2. Data Transformations and Measures Created

The raw cricket datasets required extensive cleaning, preprocessing, and transformation using Power Query to ensure analytical readiness and consistency across all dashboards.

2.1 Data Cleaning Steps

The following cleaning operations were performed:

1. Removed redundant index columns such as unnamed auto-generated columns.
2. Standardized column names to remove spaces, special characters, and numeric inconsistencies.
3. Replaced dash (-) values with zero or null values depending on analytical context.
4. Removed asterisks from high-score values while preserving “not out” information separately.
5. Fixed automatic date conversion issues in bowling best-figure columns.
6. Standardized inconsistent text values in result-type fields.
7. Converted all columns into appropriate data types including:
    - Whole number
    - Decimal number
    - Date
    - Text

2.2 Feature Engineering

Several additional analytical fields were created to improve insight generation:

1. Split the span field into:
    - Start Year
    - End Year
2. Created a **High Score Not Out Flag** to preserve dismissal context.
3. Extracted bowling performance into:
    - Wickets taken
    - Runs conceded
4. Created a unique **Match Identification Column** for tracking matches.
5. Added a **Validity Flag** to identify:
    - Tied matches
    - No-result matches

2.3 Data Integration

The batting and bowling datasets were merged using:

- Player Name
- Team Name

A **Left Outer Join** was used to ensure all player records were retained even when bowling information was missing.

The final unified **Players Table** contains:

- 518 records
- 28 columns

This integrated structure combines:

- Batting statistics
- Bowling statistics
- Fielding-related attributes

into a single analytical entity.

2.4 Data Model Design

A **Star Schema** architecture was implemented to optimize dashboard performance and analytical flexibility.

The model consists of:

Fact Tables

1. Players Table
2. Matches Table

Dimension Table

1. Teams Table

Relationships were created using standardized team-name mappings to ensure consistent filtering and aggregation across visuals.

This structure improved:

- Query performance
- Slicer interaction
- Cross-filtering efficiency
- Dashboard scalability

2.5 DAX Measures (Key Metrics Created)

To enhance analytical capability and support dashboard visualizations, several DAX measures were created.

Core Measures

Batting Average

```
Batting Average = AVERAGE(Players[Ave])
```

Used in:

- KPI cards
- Scatter plots
- Player performance comparisons

Average Strike Rate

Avg Strike Rate = AVERAGE(Players[SR])

A critical batting metric used in:

- Gauge charts
- Performance analysis
- Comparative visuals

Average Economy Rate

Avg Economy = AVERAGE(Players[Econ])

Core bowling metric where lower values indicate stronger bowling performance.

Used in:

- Bowling dashboards
- Team efficiency analysis
- Prescriptive evaluation

Total Fours

Total Fours = SUM(Players[4s])

Used in:

- Boundary analysis
- Team comparison charts
- Player attacking-strength analysis

Total Sixes

Total Sixes = SUM(Players[6s])

Represents player power-hitting ability and complements fours analysis.

Match Count

```
Match Count = COUNTROWS(Matches)
```

Used in:

- KPI cards
- Match-based calculations
- Win-related measures

Advanced Measures

Win Rate %

```
Win Rate % =
VAR Wins =
   CALCULATE(
      COUNTROWS(Matches),
      Matches[Winner] = SELECTEDVALUE(Teams[Short Name])
   )
```

Used to evaluate:

- Team effectiveness
- Seasonal consistency
- Comparative success analysis

Batting Consistency

```
Bat Consistency =
VAR Pos = SUM(Players[100]) + SUM(Players[50])
VAR Duck = SUM(Players[0])

RETURN ROUND(
   IF(Pos + Duck = 0, 0,
   Pos / (Pos + Duck)) * 100, 1)
```

Measures player reliability and consistency.

Used in:

- Player ranking

- Squad selection analysis
- Prescriptive recommendations

Strike Rate Gap to Target

```
SR Gap = [Avg Strike Rate] - 130
```

Used to identify players exceeding the benchmark strike rate of 130.

Supports:

- Aggressive batting evaluation
- Prescriptive analytics
- Team optimization

Economy Rate Gap to Target

```
Economy Gap = [Avg Economy] - 7.5
```

Used to identify efficient bowlers where negative values indicate stronger performance.

Supports:

- Bowling evaluation
- Match strategy planning
- Recruitment analysis

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3. Analytics Implementation

The dashboard implements three major forms of analytics:

1. Descriptive Analytics
2. Predictive Analytics
3. Prescriptive Analytics

3.1 Descriptive Analytics

Descriptive analytics focuses on understanding historical PSL performance using summarized statistical insights and dashboard visualizations.

The analysis includes:

1. Total runs scored by players
2. Total wickets taken by bowlers
3. Team-wise performance comparison
4. Season-wise trends across all years
5. Strike rate, batting average, and economy rate distributions

The dashboard includes KPI cards and visualizations based on:

- Batting Average
- Strike Rate
- Economy Rate
- Total Fours
- Total Sixes

These visuals summarize historical performance and support comparative analysis across teams and players.

Key outcomes include identifying:

- Top-performing batsmen
- Top-performing bowlers
- Most successful franchises
- Historical trends and patterns

3.2 Predictive Analytics

Predictive analytics focuses on forecasting future performance using historical trends and player statistics.

The analysis includes:

1. Player performance trend estimation using batting average and strike rate
2. Bowling effectiveness prediction using economy rate and strike rate
3. Team-strength forecasting based on historical win patterns
4. Season performance projection using past data trends

Predictive insights are supported using:

- Average Strike Rate
- Economy Rate
- Historical performance trends

These measures help estimate:

- Future player performance
- Team strength
- Seasonal competitiveness

This analysis helps identify likely:

- Top performers
- High-value draft picks
- Emerging talent

3.3 Prescriptive Analytics

Prescriptive analytics focuses on strategic recommendations and decision-support insights.

The analysis includes:

1. Optimal team selection based on combined batting and bowling strength
2. Identification of underused high-performing players
3. Recommendations for balanced squad composition
4. Match-strategy improvements based on historical outcomes

Prescriptive decisions are driven using benchmark-based measures including:

- SR Gap to Target (130)
- Economy Gap to Target (7.5)

These measures help identify:

- High-impact batsmen exceeding strike-rate targets
- Efficient bowlers maintaining low economy rates

This enables:

- Optimized team selection
- Tactical planning
- Performance benchmarking
- Match strategy optimization

4. Key Business Questions Addressed

The dashboard addresses the following business questions:

1. Which team has the strongest overall performance across seasons?
2. Who are the top batsmen and bowlers in PSL history?
3. Which players are the most consistent performers?
4. How does team performance change across seasons?
5. Which players contribute as both batsmen and bowlers?
6. What factors influence match outcomes most strongly?
7. Which teams perform better while chasing versus batting first?
8. Which players are high-performing but underutilized?

5. Key Insights and Findings

The analysis generated several important insights:

1. A small group of players contribute the majority of runs and wickets, indicating strong performance concentration.
2. Teams with versatile multi-skill players perform more consistently across seasons.
3. Several squad players remain underutilized despite strong efficiency metrics.
4. Economy rate is a major factor influencing success in close matches.
5. Team performance fluctuates significantly across seasons, demonstrating competitive balance within the PSL.
6. Chasing teams win slightly more matches than teams batting first.
7. Benchmark-based analysis shows that players exceeding strike-rate targets and maintaining low economy rates have significantly higher impact on match outcomes.

6. Recommendations for Management

6.1 Team Selection Strategy

1. Focus on balanced squads containing multi-skill players.
2. Prioritize bowlers with strong economy rates during pressure situations.
3. Avoid over-reliance on single star performers.

4. Use strike-rate and economy benchmarks (130 SR, 7.5 Economy) as selection criteria for high-performance players.

6.2 Player Development Strategy

1. Provide more opportunities to underutilized high-efficiency players.
2. Develop young players with strong strike-rate potential.
3. Improve consistency among mid-level performers.

6.3 Match Strategy Optimization

1. Strengthen chasing strategies due to higher historical success rates.
2. Improve death-over bowling performance.
3. Optimize batting order using strike-rate and consistency metrics.

6.4 Recruitment Strategy

1. Use predictive performance metrics for player selection.
2. Target players with combined batting and bowling abilities.
3. Prioritize consistent performers across multiple seasons.
4. Prioritize players with positive SR Gap and negative Economy Gap for data-driven recruitment decisions.

Conclusion

This project successfully transforms raw PSL cricket data into a structured analytical intelligence system using Power BI.

Through systematic:

- Data cleaning
- Data transformation
- Feature engineering
- Data modeling

- DAX measure creation

a complete semantic analytics model was developed to support:

- Descriptive analytics
- Predictive analytics
- Prescriptive analytics

The final dashboard provides actionable insights for:

- Team management
- Player evaluation
- Recruitment strategy
- Tactical decision-making

making it a complete cricket performance intelligence system powered by Power BI.